

Experiment 8 :

Implement a simple stochastic Part-of-Speech tagging algorithm using a basic probabilistic model to assign POS tags using Python.

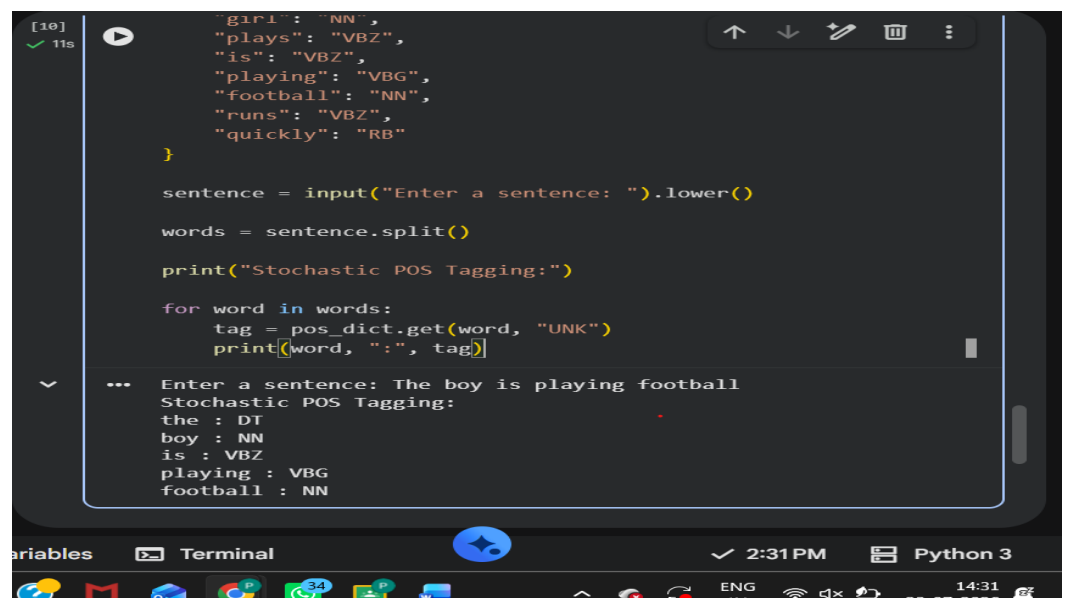
Code:

```
# Simple probabilistic dictionary

pos_dict = {
    "the": "DT",
    "boy": "NN",
    "girl": "NN",
    "plays": "VBZ",
    "is": "VBZ",
    "playing": "VBG",
    "football": "NN",
    "runs": "VBZ",
    "quickly": "RB"
}

sentence = input("Enter a sentence: ").lower()
words = sentence.split()
print("Stochastic POS Tagging:")
for word in words:
    tag = pos_dict.get(word, "UNK")
    print(word, ":", tag)
```

Output



```
[10] ✓ 11s
```

```
    "girl": "NN",
    "plays": "VBZ",
    "is": "VBZ",
    "playing": "VBG",
    "football": "NN",
    "runs": "VBZ",
    "quickly": "RB"
}

sentence = input("Enter a sentence: ").lower()
words = sentence.split()
print("Stochastic POS Tagging:")

for word in words:
    tag = pos_dict.get(word, "UNK")
    print(word, ":", tag)
```

```
... Enter a sentence: The boy is playing football
Stochastic POS Tagging:
the : DT
boy : NN
is : VBZ
playing : VBG
football : NN
```